

Invertible Ultra-Divisor Linearization on the Nonconstant Branch: Exact Orbit Preimages, Diagonal Cell Energy, and the Cross-Cell Gate

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July 2026

Abstract

TPC-98 returns the constant-phase branch at the endpoint scale, and TPC-99 rewrites the remaining conductor- q_X resonance as a multiplicative incidence problem. A literal provenance audit shows that its weight is attached to a resolved branch $\beta = (\theta, c, \kappa)$, not to a free orbit point. The native key θ fixes j_θ, σ_θ , while the actual opened divisor and moving source row vary along the resolved z -fiber. We build the multiplier linearization only after respecting this indexing.

First refine the branch absolute mass exactly into opened atoms

$$p = (\beta, z), \quad t_p = \tau_\beta + B_\beta z, \quad u_p = U_\theta(t_p), \quad m_p = M_\theta(t_p).$$

Their source identity is

$$m_p j_\theta + h_0 = \sigma_\theta u_p.$$

The branch length remains $N_\beta = \#I_\beta^*$, and its TPC-99 resonance width is $H_\beta = H_{q_X}(N_\beta)$; neither quantity is recomputed from an opened-atom cell.

On the nonconstant branch, split the opened atoms according to $q_X \mid u_p$. If $q_X \mid u_p$, then $q_X \mid m_p j_\theta + h_0$. Strong no-wrap leaves at most one orbit point for each actual moving source row. Reassembling the source inverse, both polarizations, all content and projector children, masks, and low-frequency multiplier mass gives

$$\boxed{\mathcal{L}_{K,R,X}^{q|u} \ll X^{o(1)} Q_X^2.}$$

If $q_X \nmid u_p$, the native multiplier of β satisfies

$$\omega_\beta = \varepsilon_\beta \frac{\Omega_\beta}{c_\beta} \equiv A_p(m_p j_\theta + h_0) \pmod{q_X}, \quad A_p = \varepsilon_\beta \ell_\beta v_\beta B_\beta (c_\beta u_p)^{-1}.$$

The integer provenance is $\Omega_\beta/c_\beta = \kappa_\beta \ell_\beta v_\beta (\sigma_\beta/g_\beta)$; $\ell v B/c$ need not be integral. We augment every atom by its TPC-93 child-to-source inverse and form cells by deleting only the coordinate j_θ . All other branch, z -fiber, source, content, mask, and width labels remain. Thus a cell and j recover the unique atom. In fact the retained physical values m_p, u_p, σ_p and the source identity already recover j , so these maximally literal cells are singletons. Their internal injectivity is therefore exact but vacuous; every distinct-atom collision remains in the cross-cell ledger.

The exact TPC-99 census is recovered because the opened weights satisfy $w_\beta = \sum_z w_{\beta,z}$. Its centered energy splits into an opened-atom diagonal ledger and an explicit cross-cell collision excess. Cross-cell pairs include different z 's of the same branch as well as different branches. The resulting principal, diagonal, and cross-cell estimates are sufficient gates, not theorems proved here. The finite identities are L0; the complete $q_X \mid u_p$ return and literal crosswalk are L1. No L2 fixed-shift cancellation, parity breakthrough, or prime-pair theorem is claimed.

Keywords: fixed shift; ultra divisor; no-wrap prime; native phase multiplier; affine orbit; resonance preimage; collision energy; exceptional sector.

Literal-frame convention. Every physical statement retains one prescribed nonzero integer h_0 , both matched polarizations, the original row weights, every opened divisor and exact-content child, the actual masks and support, the signed low-frequency lift, every native provenance key, and the single original global normalization. A cell below is a lossless partition of literal keys, never a quotient by equal multipliers.

1 Introduction

TPC-93 writes the literal low-frequency reconstruction term as primitive determinant- h_0 affine Möbius fibers while retaining the physical child-to-source inverse [1]. TPC-94 resolves exact content and proves that a literal additive conductor is 1 or the strong no-wrap prime q_X [2]. TPC-98 returns the conductor-one branch at $X^{o(1)}Q_X^2$ [3]. TPC-99 then writes nonconstant resonance as

$$r\omega_\beta \in E_{H_\beta}, \quad \omega_\beta = \varepsilon_\beta \frac{\Omega_\beta}{c_\beta} \in \mathbb{F}_{q_X}^\times,$$

and isolates a centered multiplier-dispersion gate [4].

The native indices impose a constraint on any further refinement. A TPC-93 key

$$\theta = (L, \gamma, \ell, j, \sigma, v, \iota)$$

already fixes j and σ . After content resolution, $\beta = (\theta, c, \kappa)$ fixes the progression and the resolved fiber I_β^* . Along that fiber,

$$t = \tau_\beta + B_\beta z,$$

the actual source data are

$$u_{\beta,z} = U_\theta(t), \quad m_{\beta,z} = M_\theta(t).$$

Thus u and m are *opened atom values*; neither is a fixed outer coordinate of β . The branch resonance width is computed before opening:

$$N_\beta = \#I_\beta^*, \quad H_\beta = H_{q_X}(N_\beta).$$

1.1 The provenance-safe route

This paper follows four steps.

1. Split the exact branch absolute mass losslessly as $w_\beta = \sum_{z \in I_\beta^*} w_{\beta,z}$, producing r -free atoms $p = (\beta, z)$.
2. Apply the TPC-93 inverse to each atom, recording its actual polarization, source row pair, opened divisor, row-gcd child, masks, content labels, and original branch width.
3. Split the atom mass by $q_X \mid u_{\beta,z}$. Return the divisible- u submass on the source ledger, where the physical row- ℓ^1 and divisor envelopes apply.
4. On $q_X \nmid u_{\beta,z}$, linearize the branch multiplier using the actual source identity and group atoms only through a deletion map whose sole deleted coordinate is j .

This order matters. Splitting by a fictitious branchwise u , or assigning the full z -fiber mass independently to every opened atom, would change the TPC-99 census.

1.2 What advances

The $q_X \mid u_{\beta,z}$ opened submass is returned at the target scale. On the complementary submass, every fine provenance cell has an affine multiplier

$$\omega = A(mj + h_0)$$

with $A, m \not\equiv 0 \pmod{q_X}$, hence only diagonal internal collisions. Because the retained source values already determine j , every such fine cell is a singleton. The complete centered energy is therefore an exact sum of a single-atom diagonal ledger and a cross-cell excess; no within-cell dispersion gain is inferred.

This is deliberately a fine decomposition. Two atoms belonging to the same β but different z 's normally have different source values $u_{\beta,z}, m_{\beta,z}$, so they lie in different cells while retaining the same ω_β . Their collision is therefore present in the cross-cell term. No false dispersion is created by atomizing the branch weight.

1.3 Organization

Section 2 constructs the opened-atom ledger and its lossless cells. Section 3 treats $q_X \mid u_{\beta,z}$. Section 4 proves the integer and finite-field formulas. Section 5 gives exact resonance preimages and collision congruences. Sections 6 and 7 reassemble the original TPC-99 census and state the remaining gates. The final sections record the finite certificate, route ledger, and claim boundary.

2 Branches, opened atoms, and lossless cells

2.1 The resolved branch is the primary key

Fix a soluble nonempty canonical branch

$$\beta = (\theta, c, \kappa), \quad \theta = (L, \gamma, \ell, j, \sigma, v, \iota).$$

Put

$$b_\beta = c_\beta \kappa_\beta, \quad g_\beta = (b_\beta, \sigma_\beta), \quad B_\beta = \frac{b_\beta}{g_\beta}. \quad (1)$$

Content resolution gives one progression

$$t_\beta(z) = \tau_\beta + B_\beta z, \quad z \in I_\beta^*.$$

Its cardinality is $N_\beta = \#I_\beta^*$. On the nonconstant resonant carrier, the TPC-99 exact resonance-window lemma gives $N_\beta \leq q_X$ for every nonzero resonant frequency, and defines the width

$$H_q(N) = \min \left\{ \frac{q-1}{2}, \left\lfloor \frac{q}{N} \right\rfloor \right\},$$

is

$$H_\beta = H_{q_X}(N_\beta). \quad (2)$$

Branches with $N_\beta > q_X$ have no nonzero resonance and are not entered into a width census. The source-level $q_X \mid u_p$ bound below does not use H_β and may be applied before this restriction. H_β denotes only this resonance width; it is not the robustness-content variable denoted H_β in the general TPC-94 normalization. On the canonical branch that earlier content equals 1. The phase slope and native multiplier are constant on this z -fiber:

$$\Omega_\beta = \ell_\beta v_\beta \sigma_\beta B_\beta, \quad \omega_\beta = \varepsilon_\beta \frac{\Omega_\beta}{c_\beta} \pmod{q_X}. \quad (3)$$

The actual source values are not constant on the fiber. For $z \in I_\beta^*$, define

$$p = (\beta, z), \quad u_p = U_\theta(t_\beta(z)), \quad m_p = M_\theta(t_\beta(z)). \quad (4)$$

Here p is an r -free opened atom. Since

$$M_\theta(t) = \ell_\theta v_\theta D_\theta(t), \quad \sigma_\theta U_\theta(t) = M_\theta(t) j_\theta + h_0,$$

every atom satisfies the exact source identity

$$m_p j_p + h_0 = \sigma_p u_p, \quad j_p = j_\theta, \quad \sigma_p = \sigma_\theta. \quad (5)$$

Different z 's of one branch generally give different u_p and m_p , while $j_p, \sigma_p, \omega_\beta, \mathbf{H}_\beta$ remain fixed.

2.2 Exact atomization of the TPC-99 weight

Let $A_{\beta,X}$ be the literal outer branch coefficient from TPC-94 and let $\mathcal{W}_{\beta,X}(z)$ contain the actual smooth, periodic, extraction, and coprimality masks. Define

$$w_{\beta,z} = |A_{\beta,X}| |\mathcal{W}_{\beta,X}(z)| \mu^2(\tilde{D}_\beta(z)) \mu^2(V_\beta(z)), \quad (6)$$

$$w_\beta = \sum_{z \in I_\beta^*} w_{\beta,z}. \quad (7)$$

Thus w_β is exactly the proof-carrying TPC-99 branch weight: in the notation of that paper,

$$w_\beta = w_{\beta,X} = |A_{\beta,X}| \mathcal{A}_{\beta,X}.$$

The refinement (7) is an equality, not a second copy of the branch mass.

For exact width H , the native census is therefore

$$\begin{aligned} a_H(\omega) &= \sum_{\substack{\beta: \mathbf{H}_\beta = H \\ \omega_\beta = \omega}} w_\beta \\ &= \sum_{\substack{p=(\beta,z): \mathbf{H}_\beta = H \\ \omega_\beta = \omega}} w_{\beta,z}. \end{aligned} \quad (8)$$

The frequency r is not part of p ; it remains in the low-window weight $b_X(r) = |\mathbf{m}_{K,X}(r)|$.

Remark 2.1 (Width is not cell occupancy). Throughout the paper,

$$\mathbf{H}_p = \mathbf{H}_\beta = H_{q_X}(\#I_\beta^*).$$

It is the resonance width of the original branch z -fiber. It is never replaced by $|J_C|$, by the number of opened atoms in a cell, or by the size of either ultra-divisor submass.

2.3 The augmented atom record

TPC-93 gives an explicit inverse from a child (θ, t) to the preprojector source atom. Augment $p = (\beta, z)$ by:

- the child coordinate $t_p = \tau_\beta + B_\beta z$ and the inverse source atom, including polarization, actual moving and opposite rows, j_p , and the opened value u_p ;
- the projector divisor v_p , interval component, actual masks, prefix, residue, smooth, and reconstruction labels;
- $c_\beta, \kappa_\beta, b_\beta, g_\beta, B_\beta$, the branch interval I_β^* , $N_\beta, \mathbf{H}_\beta$, and z ; and
- every label needed to reverse the exact-content and source reconstructions; the atom weight $w_{\beta,z}$ is attached as a coefficient payload, not used as a grouping label.

Redundant derived values such as m_p, u_p, σ_p are retained on purpose; they are provenance checks, not new multiplicities.

Definition 2.2 (Fine provenance cell). Flatten the augmented provenance record: β is not kept as an opaque token that could secretly retain j_β . Let $\text{del}_j(p)$ delete the single native coordinate j_p and its redundant copies, while deleting no independent coordinate. The attached numerical weight is not part of this projection. A fine cell C is one fiber of del_j . Write

$$P_C = \{p : \text{del}_j(p) = C\}, \quad J_C = \{j_p : p \in P_C\}.$$

Theorem 2.3 (Lossless opened-atom disintegration). *Assume the TPC-93 child-to-source inverse and the TPC-94 exact content reconstruction. Then:*

- (i) (C, j_p) reconstructs the unique pair (β, z) , its r -free source atom, and its weight $w_{\beta,z}$;
- (ii) in fact C already determines

$$j_p = \frac{\sigma_p u_p - h_0}{m_p},$$

so every nonempty fine cell P_C is a singleton;

- (iii) all $p \in P_C$ have common values $m_C, u_C, \ell_C, v_C, c_C, B_C, \varepsilon_C, H_C$; and
- (iv) summing the cells gives exactly the second line of (8).

Proof. The flattened augmented record contains every coordinate of (β, z) , its TPC-93 inverse source record, and every reconstruction label. It contains no opaque branch identifier. The map del_j removes only the native j coordinate and direct stored copies, so restoring that coordinate restores the original record and its attached coefficient. This proves (i). The retained physical coordinates m_p, u_p, σ_p remain. Their source identity (5) determines the displayed integer j_p , proving (ii). Every quantity in (iii) remains in the deletion record, either as a native coordinate or an explicitly retained source value. Finally, the cells form a disjoint partition of the opened atoms, so (iv) is just finite reindexing of (8). $\square$

Remark 2.4 (No claim that θ fixes u). The cell fixes the *opened source value* $u_C = U_\theta(t_p)$ only after z has been opened and the source inverse recorded. The branch β , and in particular θ , does not fix u or m along its full resolved fiber.

Remark 2.5 (The fine-cell injection is deliberately vacuous). The singleton conclusion is not an equidistribution statement. It is the unavoidable price of retaining the complete source inverse while deleting only the displayed j -coordinate: the determinant identity algebraically recovers that coordinate. Consequently all collisions between distinct opened atoms, including same- β different- z collisions, occur in the cross-cell ledger.

2.4 Strong no-wrap consequences

The strengthened prime of TPC-98 may be taken, still with $q_X \asymp Q_X$, larger than every actual source row, every relevant row difference, and the orbit diameter. On active canonical support,

$$0 < m_p, \ell_p, v_p, c_\beta, B_\beta, b_\beta < q_X, \quad \text{diam}(\mathcal{J}_X) < q_X. \quad (9)$$

Hence $m_p, \ell_p, v_p, c_\beta, B_\beta$ are invertible modulo q_X . On the nonconstant branch,

$$q_X \nmid \Omega_\beta \iff q_X \nmid \sigma_\beta, \quad \omega_\beta \in \mathbb{F}_{q_X}^\times. \quad (10)$$

The equivalence uses the literal bound $B_\beta < q_X$ and the nonvanishing of ℓ_β, v_β modulo q_X [3].

3 The opened ultra-value split

The split is made after (7), at the atom level:

$$w_\beta = \sum_{\substack{z \in I_\beta^* \\ q_X \mid u_{\beta,z}}} w_{\beta,z} + \sum_{\substack{z \in I_\beta^* \\ q_X \nmid u_{\beta,z}}} w_{\beta,z}. \quad (11)$$

This is a partition of one branch absolute mass. It does not assert that u is an outer key of β .

Theorem 3.1 (One orbit point for a fixed actual source row). *Fix an actual moving source row m . Among orbit points $j \in \mathcal{J}_X$, at most one orbit coordinate can occur for opened atoms satisfying*

$$m_p = m, \quad q_X \mid u_p.$$

Proof. For such an atom, (5) gives

$$q_X \mid u_p \implies m j_p + h_0 \equiv 0 \pmod{q_X}.$$

Since $q_X \nmid m$, this fixes one residue class for j_p . The actual orbit support has diameter below q_X , hence contains at most one representative. The theorem does not rule out several literal children over that same reconstructed source point; those children are restored with their recorded absolute multiplicities below. $\square$

Remark 3.2. The theorem does not fix β and then vary j : a branch already fixes j_θ . It first uses the TPC-93 inverse

$$(\beta, z) \mapsto (\text{polarization, actual source rows, } j_p, u_p)$$

and then counts the reconstructed source atoms by their actual moving row.

3.1 Restoring the physical absolute ledger

The imported physical estimates are:

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}| \ll_{\mathcal{D}} Q_X, \quad i = 1, 2, \quad (12)$$

$$\sum_{\substack{u \mid N \\ T_X < u \leq U_{0,X}}} |\mu(u)| \log u \leq X^{o(1)}, \quad (13)$$

$$\sum_{0 < d_{q_X}(r,0) \leq R_X} |\mathfrak{m}_{K,X}(r)| \ll_K 1. \quad (14)$$

The opposite-prefix envelope, row-gcd projector mass, exact-content absolute multiplicity, finite separation, interval split, and local-mask reconstruction each have total mass $X^{o(1)}$ on the literal packet [1, 3].

Theorem 3.3 (Complete return of the opened $q_X \mid u_p$ submass). *Let $\mathcal{L}_{K,R,X}^{q|u}$ be the part of the nonconstant absolute low-window carrier obtained from the first sum in (11). Under the imported literal source-to-child inverse and the estimates above,*

$$|\mathcal{L}_{K,R,X}^{q|u}| \ll_{K,h_0,\mathcal{D}} X^{o(1)} Q_X^2. \quad (15)$$

Proof. Apply the TPC-93 inverse to every opened atom. On the left-ultra polarization, theorem 3.1 leaves at most one orbit point for each actual moving row. At that point, summing all actual opened values $u_p = U_\theta(t_p)$ divisible by q_X is bounded by the full envelope (13). The opposite prefix and bounded local factors cost $X^{o(1)}$. Thus the source shell is

$$\ll X^{o(1)} \left(\sum_{\alpha} |\gamma_{\alpha}^{(1)}| \right) \left(\sum_{\gamma} |\gamma_{\gamma}^{(2)}| \right) \ll X^{o(1)} Q_X^2.$$

The right-ultra polarization is symmetric.

Now restore the row-gcd children through their projector weights and overcount the exact-content, coprimality, interval, residue, smooth, and mask children by their recorded soft absolute multiplicities. This reconstructs the atom weights in (6); it does not attach the full w_β to each z . Finally use (14) for all low frequencies and apply the original global normalization once. $\square$

Remark 3.4 (Disjoint from the constant return). The ambient branch has $q_X \nmid \sigma_\beta$. TPC-98 treats $q_X \mid \sigma_\beta$, whereas [theorem 3.3](#) treats atom values $q_X \mid U_\theta(t_\beta(z))$. The two subledgers are disjoint.

4 Atomwise invertible multiplier linearization

4.1 Integer provenance at the branch level

Lemma 4.1 (Canonical integer factorization). *Write $\sigma_\beta = g_\beta \sigma_{\beta,0}$. Then*

$$\frac{\Omega_\beta}{c_\beta} = \kappa_\beta \ell_\beta v_\beta \sigma_{\beta,0} = \kappa_\beta \ell_\beta v_\beta \frac{\sigma_\beta}{g_\beta} \in \mathbb{Z}. \quad (16)$$

Proof. Since $b_\beta = g_\beta B_\beta = c_\beta \kappa_\beta$,

$$\Omega_\beta = \ell_\beta v_\beta (g_\beta \sigma_{\beta,0}) B_\beta = c_\beta \kappa_\beta \ell_\beta v_\beta \sigma_{\beta,0}.$$

$\square$

Example 4.2 (No integer quotient $\ell v B/c$). *Take*

$$c = 2, \quad \kappa = 1, \quad b = 2, \quad \sigma = 2, \quad g = 2, \quad B = 1, \quad \ell = v = 1.$$

Then $\Omega/c = 1$, but $c \nmid \ell v B$. Therefore $(\ell v B/c)\sigma$ has the correct rational value but does not exhibit an integer coefficient. Division by c below is a finite-field inversion, not this extra integer divisibility.

4.2 The actual opened value is the invertible divisor

Theorem 4.3 (Atomwise affine multiplier formula). *Let $p = (\beta, z)$ be a nonconstant opened atom with $q_X \nmid u_p$. Define*

$$A_p = \varepsilon_\beta \ell_\beta v_\beta B_\beta (c_\beta u_p)^{-1} \in \mathbb{F}_{q_X}^\times. \quad (17)$$

Then

$$\boxed{\omega_\beta = A_p (m_p j_p + h_0) \quad \text{in } \mathbb{F}_{q_X}^\times.} \quad (18)$$

Equivalently,

$$A_p = \varepsilon_\beta \kappa_\beta \ell_\beta v_\beta (u_p g_\beta)^{-1}. \quad (19)$$

Proof. All factors in (17) are nonzero modulo q_X . Reduce the branch multiplier and use the atom source identity:

$$\begin{aligned} \omega_\beta &\equiv \varepsilon_\beta \ell_\beta v_\beta B_\beta c_\beta^{-1} \sigma_\beta \\ &\equiv \varepsilon_\beta \ell_\beta v_\beta B_\beta (c_\beta u_p)^{-1} (m_p j_p + h_0). \end{aligned}$$

This proves (18). The identity $B_\beta g_\beta = c_\beta \kappa_\beta$ gives (19). $\square$

Remark 4.4 (Dependence on z). The left side ω_β is constant for $z \in I_\beta^*$. The coefficient A_p and source row m_p generally vary with z , because

$$u_p = U_\theta(\tau_\beta + B_\beta z), \quad m_p = M_\theta(\tau_\beta + B_\beta z).$$

No theorem here treats u or m as fixed on β .

Theorem 4.5 (Fine-cell injectivity). *Let C be a fine provenance cell consisting of atoms with $q_X \nmid u_p$. Then the common values in [theorem 2.3](#) define*

$$A_C = \varepsilon_C \ell_C v_C B_C (c_C u_C)^{-1},$$

and

$$p \longmapsto \omega_{\beta(p)} = A_C(m_C j_p + h_0) \quad (20)$$

is injective on P_C .

Proof. By [theorem 2.3\(ii\)](#), P_C is already a singleton. Equivalently, one may verify the affine implication directly as follows. If two atoms $p, p' \in P_C$ have equal multipliers, then

$$A_C m_C (j_p - j_{p'}) = 0 \pmod{q_X}.$$

Since $A_C m_C \neq 0$, the two orbit coordinates are congruent modulo q_X . They lie in the true orbit window of diameter below q_X , so $j_p = j_{p'}$. By [theorem 2.3\(i\)](#), the common cell record and this coordinate reconstruct the same (β, z) . Hence $p = p'$. $\square$

Remark 4.6. This theorem records compatibility of the affine formula with the lossless provenance partition. It supplies no nontrivial within-cell dispersion because the maximally literal cells are singletons.

Remark 4.7 (Same branch, different atoms). If $z \neq z'$ belong to one branch β , then both atoms have the same ω_β , but their opened source values generally differ. They therefore lie in different fine cells, and their collision is retained as a cross-cell term. Fine-cell injectivity does not convert a branchwise point mass into dispersed mass.

5 Exact preimages with the inherited branch width

Let $q = q_X$ and

$$E_H = \{\pm 1, \dots, \pm H\} \subset \mathbb{F}_q^\times.$$

For a branch β of length $N_\beta \leq q$, TPC-99 proves

$$\beta \text{ is resonant at } r \iff r\omega_\beta \in E_{H_\beta}, \quad H_\beta = H_q(N_\beta).$$

Every atom inherits this same test.

By [theorem 2.3](#), every $j \in J_C$ corresponds to a unique atom $p_C(j) \in P_C$. Write

$$\omega_C(j) = \omega_{\beta(p_C(j))}.$$

Theorem 5.1 (Exact resonance preimage in a fine cell). *Let C be a $q \nmid u_p$ cell, fix $r \in \mathbb{F}_q^\times$, and let*

$$H_C = H_{\beta(p)} = H_q(N_{\beta(p)}) \quad (p \in P_C).$$

Then

$$\{j \in J_C : r\omega_C(j) \in E_{H_C}\} = J_C \cap \bigcup_{s \in E_{H_C}} \left\{ m_C^{-1}((rA_C)^{-1}s - h_0) \right\} \pmod{q}. \quad (21)$$

Consequently,

$$\#\{j \in J_C : r\omega_C(j) \in E_{H_C}\} \leq \min\{|J_C|, 2H_C\}. \quad (22)$$

Proof. Use (20) and solve

$$rA_C(m_C j + h_0) = s \pmod{q}$$

for each $s \in E_{H_C}$. All coefficients are invertible and distinct s 's give distinct residues. Each residue meets the no-wrap orbit window in at most one integer. $\square$

Remark 5.2. The bound uses $2H_C = 2H_q(N_\beta)$, not $2H_q(|J_C|)$. Atomization changes neither the phase length nor the resonance threshold of the original branch.

Corollary 5.3 (Exact multiplier preimage). *For $y \in \mathbb{F}_q^\times$,*

$$\{j \in J_C : \omega_C(j) = y\} = J_C \cap \{m_C^{-1}(A_C^{-1}y - h_0)\} \pmod{q}, \quad (23)$$

and this set has cardinality zero or one.

Proposition 5.4 (Exact cross-cell collision). *For fine cells C, D and atoms $p_C(j), p_D(k)$,*

$$\omega_C(j) = \omega_D(k) \iff A_C m_C j - A_D m_D k \equiv h_0(A_D - A_C) \pmod{q}. \quad (24)$$

For fixed j , at most one $k \in J_D$ solves this congruence, and vice versa. Thus

$$\#\{(j, k) \in J_C \times J_D : \omega_C(j) = \omega_D(k)\} \leq \min\{|J_C|, |J_D|\}. \quad (25)$$

Proof. Expand

$$A_C(m_C j + h_0) = A_D(m_D k + h_0)$$

and rearrange. Since $A_D m_D \neq 0$, one $k \pmod{q}$ is possible; no-wrap gives at most one integer in J_D . Exchange the cells for the symmetric count. $\square$

Remark 5.5 (Fine cells are singletons). Because the deletion record retains every coordinate other than j , including σ, z and the actual source values, the source identity reconstructs j and every fine cell is a singleton. Thus the preimage and partner bounds are exact but formally trivial on these cells. The result is a lossless collision identity, not a claim that a large set of orbit points has already been equidistributed.

6 Exact recovery of branch mass and collision energy

Fix H , and retain only opened atoms with $q_X \nmid u_p$. Let $\mathcal{C}_H$ be their fine cells of inherited width $H_\beta = H$. By lossless reconstruction, for each $j \in J_C$ there is a unique

$$p_C(j) = (\beta_C(j), z_C(j)).$$

Define the opened coefficient

$$w_C(j) = w_{\beta_C(j), z_C(j)} \quad (26)$$

using (6), and put

$$a_{H,C}^{\text{inv}}(\omega) = \sum_{j \in J_C} w_C(j) \mathbf{1}_{\{\omega_C(j)=\omega\}}, \quad a_H^{\text{inv}} = \sum_{C \in \mathcal{C}_H} a_{H,C}^{\text{inv}}. \quad (27)$$

Proposition 6.1 (Exact return to the branch census). *One has*

$$a_H^{\text{inv}}(\omega) = \sum_{\substack{\beta: H_\beta=H \\ \omega_\beta=\omega}} \sum_{\substack{z \in I_\beta^* \\ q_X \nmid u_{\beta,z}}} w_{\beta,z}. \quad (28)$$

Together with the analogous $q_X \mid u_{\beta,z}$ sum, this recovers the exact TPC-99 census

$$a_H(\omega) = \sum_{\substack{\beta: H_\beta=H \\ \omega_\beta=\omega}} w_\beta.$$

Proof. The fine cells partition the r -free opened atoms and (C, j) reconstructs (β, z) . Reindex the finite sum in (27), then use (7). Every atom weight appears exactly once. $\square$

Remark 6.2 (Why the atom coefficient is forced). Replacing $w_C(j)$ by the full branch mass w_β would repeat that mass once for every z . Replacing w_β by a cell-count normalization would alter the TPC-99 carrier. The only lossless positive refinement is the opened coefficient (6), followed by exact re-summation.

Put

$$M_C = \sum_{j \in J_C} w_C(j), \quad M_H = \sum_C M_C.$$

Theorem 6.3 (Exact opened-atom collision energy). *For every inherited width H ,*

$$\begin{aligned} \|a_H^{\text{inv}}\|_2^2 &= \sum_{C \in \mathcal{C}_H} \sum_{j \in J_C} w_C(j)^2 \\ &\quad + \sum_{\substack{C, D \in \mathcal{C}_H \\ C \neq D}} \sum_{\substack{j \in J_C, k \in J_D \\ \omega_C(j) = \omega_D(k)}} w_C(j) w_D(k). \end{aligned} \quad (29)$$

All internal cell collisions are diagonal.

Proof. Expand $\sum_\omega a_H^{\text{inv}}(\omega)^2$. Inside one cell, [theorem 4.5](#) makes equality of multipliers equivalent to equality of atoms. Distinct-cell terms remain exactly under (24). $\square$

6.1 Centered decomposition

Center on $\mathbb{F}_q^\times$:

$$(a_H^{\text{inv}})^\circ = a_H^{\text{inv}} - \frac{M_H}{q-1} \mathbf{1}.$$

Define

$$\mathfrak{D}_H^\circ = \sum_C \left\{ \sum_{j \in J_C} w_C(j)^2 - \frac{M_C^2}{q-1} \right\}, \quad (30)$$

$$\mathfrak{X}_H^\circ = \sum_{C \neq D} \left\{ \sum_{\substack{j \in J_C, k \in J_D \\ \omega_C(j) = \omega_D(k)}} w_C(j) w_D(k) - \frac{M_C M_D}{q-1} \right\}. \quad (31)$$

Corollary 6.4 (Exact centered energy).

$$\|(a_H^{\text{inv}})^\circ\|_2^2 = \mathfrak{D}_H^\circ + \mathfrak{X}_H^\circ, \quad (32)$$

and $\mathfrak{D}_H^\circ \geq 0$.

Proof. Subtract $M_H^2/(q-1)$ from (29) and split

$$M_H^2 = \sum_C M_C^2 + \sum_{C \neq D} M_C M_D.$$

Fine-cell injectivity gives $|J_C| \leq q-1$; hence

$$M_C^2 \leq |J_C| \sum_j w_C(j)^2 \leq (q-1) \sum_j w_C(j)^2.$$

$\square$

Remark 6.5 (Singleton form of the diagonal gate). By [theorem 2.3\(ii\)](#), every C contains one opened atom $p = (\beta, z)$. Therefore

$$\mathfrak{D}_H^\circ = \frac{q-2}{q-1} \sum_{\substack{p=(\beta, z): \mathbf{H}_\beta = H \\ q \nmid u_p}} w_{\beta, z}^2. \quad (33)$$

Thus the diagonal gate is exactly an opened-atom weight second-moment problem, not a consequence of multiplier dispersion. Every collision between distinct atoms remains in $\mathfrak{X}_H^\circ$.

Remark 6.6 (What the cross term contains). $\mathfrak{X}_H^\circ$ includes:

- (i) atoms from different branches;
- (ii) different content or projector children of one source atom; and
- (iii) different z 's of the same branch β .

In case (iii), the multipliers are identically equal because they are both ω_β . The exact branchwise concentration is therefore retained, not disguised as cellwise dispersion.

Proposition 6.7 (Exact cell form of the incidence carrier). *For any frequency weight b ,*

$$\sum_{\omega, r} a_H^{\text{inv}}(\omega) b(r) \mathbf{1}_{E_H}(r\omega) = \sum_C \sum_{j \in J_C} w_C(j) \sum_r b(r) \mathbf{1}_{E_H}(rA_C(m_C j + h_0)). \quad (34)$$

For fixed C, r , the j 's are exactly the preimages in (21).

Proof. Insert (27) and (20), then interchange finite sums. The width is $H = H_\beta$ throughout. $\square$

7 The corrected TPC-99 crosswalk

TPC-99 proves for the full branch census

$$\mathcal{I}_H(a_H, b_X) = \frac{2H}{q_X - 1} \|a_H\|_1 \|b_X\|_1 + \mathcal{R}_H, \quad (35)$$

$$|\mathcal{R}_H| \ll_K \sqrt{q_X} \log q_X \|a_H^\circ\|_2, \quad (36)$$

with $\|b_X\|_1 + \|b_X\|_2 \ll_K 1$ [4]. The exact atomization gives

$$a_H = a_H^{q|u} + a_H^{\text{inv}}.$$

The positive incidence form is linear in a_H . The first term is returned directly by [theorem 3.3](#); no centered-energy estimate for it is needed.

Theorem 7.1 (Corrected sufficient gates for the invertible submass). *Assume the literal atomization and source inverse in [theorem 2.3](#). After the $q_X \mid u_p$ return, the remaining nonconstant resonant carrier is $O(X^{o(1)} Q_X^2)$ if*

$$\sum_H \frac{H}{q_X} M_H \ll X^{o(1)} Q_X^2, \quad (37)$$

$$\sum_H \sqrt{\mathfrak{D}_H^\circ} \ll X^{o(1)} Q_X^{3/2}, \quad (38)$$

$$\sum_H \sqrt{|\mathfrak{X}_H^\circ|} \ll X^{o(1)} Q_X^{3/2}. \quad (39)$$

Here every H is $H_q(N_\beta)$, and every mass in $M_H, \mathfrak{D}_H^\circ, \mathfrak{X}_H^\circ$ is the opened coefficient (6).

Proof. Apply (35) to a_H^{inv} . The principal term is controlled by (37). From (32),

$$\|(a_H^{\text{inv}})^\circ\|_2 \leq \sqrt{\mathfrak{D}_H^\circ} + \sqrt{|\mathfrak{X}_H^\circ|}.$$

The last two gates therefore give

$$\sum_H \|(a_H^{\text{inv}})^\circ\|_2 \ll X^{o(1)} Q_X^{3/2}.$$

Since $q_X \asymp Q_X$, the factor in (36) returns $O(X^{o(1)} Q_X^2)$. Add the already controlled $q_X \mid u_p$ atom submass. $\square$

Remark 7.2 (No diagonal theorem is hidden). Fine-cell injectivity identifies the diagonal term but does not bound its atom weights. In particular, splitting w_β into $w_{\beta,z}$ does not reduce the original point mass at ω_β : all cross terms between its z -atoms reappear in $\mathfrak{X}_H^\circ$.

Remark 7.3 (Sufficient positive route only). Neither (37), (38), nor (39) is proved here. A proof-carrying polynomial obstruction in the corresponding termwise bounds would stop this sufficient principal-plus-absolute-remainder implementation, not the original signed Möbius sum, which retains phase constants and cancellation across frequencies and branches.

8 Exact provenance regression

The script `experiments/tpc100_certificate.py` models the literal order

$$\beta = (\theta, c, \kappa) \longrightarrow (\beta, z) \longrightarrow \text{source inverse} \longrightarrow \text{del}_j \text{ cells.}$$

It constructs 4,096 resolved branches, 6,851 opened atoms, and 6,851 fine cells over small no-wrap primes. Using exact integer and rational arithmetic, it checks:

1. the affine determinant identity and the source equality $m_{\beta,z}j_\theta + h_0 = \sigma_\theta u_{\beta,z}$;
2. the exact mass equality $w_\beta = \sum_z w_{\beta,z}$ and equality of the branch and atom multiplier censuses;
3. reconstruction of (β, z) and its r -free source from the fine cell together with j , and the stronger fact that the retained physical source coordinates force every fine cell to be a singleton;
4. inheritance of $H_q(N_\beta)$, including explicit examples where it differs from $H_q(|J_C|)$;
5. both finite-field coefficients, atomwise linearization, and fine-cell injectivity;
6. forced cross-cell collisions between different z 's of one branch;
7. exact resonance preimages, cross-cell congruences and partner counts; and
8. uncentered and centered opened-atom energy identities.

The certificate completes 208,811 exact checks, including both polarization signs. Its SHA-256 hash is

EF308431ED63AC09D93B3E45C4B8915FDD851D7C32237AF192CB48DAA9A36E4F.

No randomness or floating-point comparison is used. The certificate verifies finite provenance algebra only; it is not evidence for any asymptotic gate.

9 Route and exponent ledger

The provenance-corrected route is

literal object	proved status	remaining input
$q_X \mid \Omega_\beta$	$O(X^{o(1)}Q_X^2)$ in TPC-98	none here
$q_X \nmid \Omega_\beta, q_X \mid U_\theta(t_\beta(z))$	$O(X^{o(1)}Q_X^2)$ here	none here
$q_X \nmid \Omega_\beta, q_X \nmid U_\theta(t_\beta(z))$	atomwise $\omega = A(mj + h_0)$	three explicit gates
nonconstant generic fibers	literal export only	signed two-form cancellation.

The exact order of indices is now part of the theorem:

$$\beta \longrightarrow \{(\beta, z) : z \in I_\beta^*\} \longrightarrow \text{TPC-93 source inverse} \longrightarrow \text{del}_j \text{ cells.}$$

The reverse sum first restores the opened atoms, then $w_\beta = \sum_z w_{\beta,z}$, and finally the TPC-99 census. One may not start with a branchwise fixed u or m .

The next positive inputs are therefore:

1. the principal atom-mass gate (37);
2. an opened diagonal-weight ledger (38); and
3. the centered cross-cell estimate (39), including the forced collisions between different z 's of one branch.

Alternatively one must return to the signed branch sum before forming a positive census.

No polynomial exponent is spent by the finite reindexing. The determinant compatibility

$$\lambda_D \leq 2\eta_Z$$

and the strict physical ledger

$$\Lambda_{\text{phys}} < \frac{1}{400}$$

remain independent. The $\sqrt{q_X} \log q_X$ factor is charged only if the TPC-99 positive incidence route is used.

Remark 9.1 (Sharper stop rule after the audit). Same- β , different- z atoms have identical multipliers. Moreover the maximally literal cells are singletons, because their retained source data already recover j . Consequently atomwise linearization alone cannot prove branchwise dispersion. Any future argument must quantitatively control those forced cross-cell collisions, find a genuinely coarser lossless structure, or retain signed z -fiber cancellation. Ignoring them is now an explicit provenance failure.

10 Claim boundary

Unconditional finite identities

- A branch β fixes j_θ, σ_θ and its phase multiplier, while $u_{\beta,z} = U_\theta(\tau_\beta + B_\beta z)$ and $m_{\beta,z} = M_\theta(\tau_\beta + B_\beta z)$ vary with z .
- The branch multiplier has the integer factorization

$$\Omega_\beta / c_\beta = \kappa_\beta \ell_\beta v_\beta (\sigma_\beta / g_\beta).$$

The stronger divisibility $c \mid \ell v B$ is not inferred.

- For every opened atom with $q_X \nmid u_{\beta,z}$,

$$\omega_\beta = A_{\beta,z}(m_{\beta,z} j_\theta + h_0) \pmod{q_X}.$$

- Exact resonance preimages use $H_\beta = H_q(\#I_\beta^*)$, never cell occupancy.
- Every maximally literal fine cell is a singleton because its retained source data recover j . Cross-cell collisions satisfy (24).
- The exact atom weights reassemble as $w_\beta = \sum_z w_{\beta,z}$, recovering the TPC-99 census and centered-energy decomposition without duplication.

These are L0 statements once the upstream native formulas and source inverse are supplied.

Conditional literal return

Assuming the TPC-93–TPC-98 lossless source reconstruction, row- ℓ^1 bounds, divisor and projector envelopes, actual masks, and one global normalization, the opened nonconstant submass

$$q_X \mid U_\theta(\tau_\beta + B_\beta z)$$

is $O(X^{o(1)} Q_X^2)$. This one-subledger return and the exact TPC-99 atom crosswalk are L1 progress.

The construction retains:

- the complete $\beta = (\theta, c, \kappa)$ and z labels;
- $b, c, \kappa, g, B, I_\beta^*, N_\beta, H_\beta$;
- the TPC-93 child-to-source inverse, actual source row pair, opened value $u_{\beta,z}$, row-gcd child, and masks;
- both polarizations and ε_θ ;
- the external frequency ledger, which separately retains the least-absolute lift $\tilde{r}$, signed reconstruction multiplier, and modulus cq_X ; and
- the original branch weight and global normalization.

Not proved

- The principal, diagonal, and cross-cell gates (37)–(39) are not proved.
- Fine-cell injectivity is vacuous, not branchwise or global multiplier dispersion. Different z 's of the same β are forced multiplier collisions.
- The positive carrier is not identified with the original signed z -fiber sum.
- No complete short- or generic-sector bound, affine Möbius cancellation theorem, determinant-energy lower bound, TPC-18 cover, parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime conclusion is proved.
- Nothing is specialized to $h_0 = 2$.

The audit weakens the interpretation of cellwise injectivity but makes the literal crosswalk correct. The paper removes one opened submass and exposes the true remaining collision gate; it does not claim L2 progress.

References

- [1] Liang Wang. Literal low-window affine export: Content-factor extraction, resonance sectors, and fixed-determinant preservation, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-93-literal-low-window-affine-export. Preprint.
- [2] Liang Wang. Exact-content progression normalization and resonance: Primitive determinants, rational conductors, and a complete exceptional-sector ledger, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-94-exact-content-resonance-ledger. Preprint.
- [3] Liang Wang. The literal constant-phase sector at the strong no-wrap prime: Orbit sparsity and an endpoint-scale literal bound, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-98-long-constant-phase-orbit-sparsity. Preprint.
- [4] Liang Wang. Multiplicative resonance incidence at the no-wrap prime: Exact character spectrum, parity nullspace, and the native dispersion gate, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-99-multiplicative-resonance-incidence. Preprint.